

PROJECT BLUEPRINT · v1.0

Hindustani Jyotish

हिन्दुस्तानी ज्योतिष

A two-app Vedic astrology platform — User & Astrologer

Chat · Call · Video · Live Streaming · Kundli · Wallet

Project	Hindustani Jyotish — Vedic Astrology Platform
Apps	User App + Astrologer App (Android · iOS · Web)
Stack	Flutter (BLoC · GetIt · go_router) + FastAPI + PostgreSQL
Auth	Google Sign-In + Biometric (FaceID / Fingerprint) + OTP
Owner	Jaikalki Technology Pvt. Ltd.
Date	30 May 2026
Status	Draft — Pre-Development Reference

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1. Executive Summary

> **What is Hindustani Jyotish?**
>
> A mobile-first Vedic astrology platform with two paired apps. The User App lets seekers find verified astrologers, generate Kundli charts, consult via chat/audio/video, watch live streams, and pay through a wallet. The Astrologer App lets verified jyotishis go live, attend incoming requests, track real-time earnings, and withdraw to bank. Both apps share one cosmic-premium identity: Deep Indigo + Saffron + Copper, with Devanagari typography.

Core Value Propositions

- **For Seekers (User App):** Talk to verified Vedic astrologers in minutes — chat from ₹7/msg, call from ₹20/min, video from ₹25/min. Generate detailed Kundli, daily horoscope, planet positions, love compatibility, and birthday number reports. Watch live rashifal streams.
- **For Astrologers (Astrologer App):** Single dashboard for all consultations. Toggle online/offline. Accept or decline incoming chat/call/video requests with a 30-second countdown. Live-stream to followers. Real-time earnings tracker with one-tap withdrawal.
- **For Platform (Jaikalki):** Commission on every consultation (configurable 20–30%). Super-chat & gifting revenue during live streams. Premium paid reports (Kundli ₹99, Planet Position ₹79, etc).

Success Metrics (North-Star)

Metric	Target — Month 6	Target — Month 12
Verified Astrologers	50	200
Daily Active Users	5,000	25,000
Daily Consultations	500	3,000
Avg Consultation Length	8 min	12 min
Monthly Revenue	₹8 L	₹50 L

2. Brand Identity Recap

The brand is built around a single visual idea — the diya (lamp of inner knowledge) sitting at the centre of an orbital ring of nakshatra dots, crowned by a guiding star. This translates into a logo system that works equally on dark indigo splash screens and on ivory marketing collateral.

Colour System

Token	Hex	Role
Indigo Void	#0B0B1E	Primary canvas — every screen background
Indigo Mid	#2D2866	Card surfaces, headers
Saffron	#F4811F	Primary action — CTAs only, never decoration
Copper	#C77B4A	Astrologer-side accent, premium states
Gold	#E8B547	Ratings, wallet balance, paid reports
Ivory	#FBF6EE	Primary text on dark, light-mode surfaces
Success Green	#34D399	Earnings, "you earned" states, online indicators
Live Red	#FF4D6D	Live stream badge, end-call, danger CTAs

Typography

Family	Role	Where
Fraunces (serif)	Display & section titles	App splash, hero titles, screen headers
Marcellus (serif)	Deco accents	Numbers, large stats, splash secondary text
Plus Jakarta Sans	Body & UI	All paragraph text, buttons, inputs, labels
Noto Sans Devanagari	Hindi/Sanskrit	हिन्दुस्तानी, नमस्ते, app tagline
JetBrains Mono	Meta	Status, timestamps, OTP, transaction IDs

3. Technology Stack

Frontend (Both Apps)

Layer	Library	Purpose
Framework	Flutter 3.24+	Cross-platform Android / iOS / Web
State Management	flutter_bloc (Cubit + Bloc)	Predictable state per feature
DI	get_it + injectable	Service locator pattern
Routing	go_router 14+	Declarative, deep-linkable navigation
Network	dio + retrofit	Type-safe REST client, interceptors, retry
Local Storage	flutter_secure_storage + hive	Tokens (secure) + offline cache (hive)
Auth — Google	google_sign_in 6+	Google OAuth flow
Auth — Biometric	local_auth 2.3+	Fingerprint, FaceID, iris
Real-Time A/V	agora_rtc_engine 6+	Chat, audio call, video call, live broadcast
Push Notifications	firebase_messaging	Incoming request alerts, live-start alerts
Payments	razorpay_flutter	Wallet recharge, super-chat purchase
Image / Asset	cached_network_image	Astrologer photos with placeholder & cache
Date / Time	intl + timezone	Birth-time correctness, IST display

Backend

Layer	Tech	Purpose
API Framework	FastAPI (Python 3.12)	Async REST, OpenAPI auto-docs
Database	PostgreSQL 16	Primary store — users, consultations, transactions
ORM	SQLAlchemy 2.0 + Alembic	Models, migrations, typed queries
Caching	Redis	Session, online-astrologer pool, rate-limit
Object Storage	AWS S3 / Cloudflare R2	Profile photos, kundli PDFs, KYC docs
Auth Layer	OAuth2 (Google) + JWT + bcrypt OTP	Multi-method auth, refresh tokens
Background Jobs	Celery + Redis broker	Wallet settlement, payout, email/SMS
SMS Gateway	MSG91 / Twilio	OTP, transactional alerts
Email	SendGrid	Receipts, kundli reports, notifications

Layer	Tech	Purpose
Live A/V Server	Agora SD-RTN (managed)	No self-hosting
Push	Firebase Cloud Messaging	Cross-platform push
Monitoring	Sentry + Grafana + Prometheus	Errors, metrics, alerts

4. System Architecture

High-Level Flow

Both Flutter apps talk only to the FastAPI gateway. The gateway authenticates the request via JWT (after Google or biometric unwraps the secure-storage token), dispatches to internal services, and proxies real-time A/V tokens for Agora.

Component Map

Tier	Components	Talks To
CLIENT	Flutter User App, Flutter Astrologer App	API Gateway, Agora SDK, FCM, Razorpay
GATEWAY	FastAPI (auth, routing, validation, rate-limit)	Auth Service, App Services, Redis
AUTH	Google OAuth, OTP service, JWT issuer, biometric token vault	PostgreSQL (users), Redis (sessions), MSG91 (SMS)
CORE	User svc, Astrologer svc, Consultation svc, Wallet svc, Live svc, Kundli svc	PostgreSQL, S3, Agora, Razorpay
JOBS	Celery workers — wallet settlement, payout, kundli PDF, notifications	PostgreSQL, SendGrid, MSG91, FCM
DATA	PostgreSQL (OLTP), Redis (cache + queue), S3 (objects)	—

Real-Time A/V Flow (Agora)

1. Client requests channel + RTC token from FastAPI endpoint (authenticated).
2. FastAPI signs an Agora RTC token (HMAC) with the user UID and channel name, returns it.
3. Client joins the Agora channel directly — media never traverses our servers.
4. On call end, client posts duration + cost to FastAPI; wallet service debits user, credits astrologer (minus commission).

5. Authentication Strategy

Every phase, every screen, every protected route respects this layered model. Three entry methods, one unified session.

5.1 Three Authentication Methods

A. Google Sign-In (primary, fastest)

One-tap account creation/login using the user's Google account. Both apps offer this on the very first screen.

- **Flutter SDK:** google_sign_in 6.x
- **Backend:** Verify Google ID token via Google public keys → upsert user → issue our JWT pair (access 15 min + refresh 30 days)
- **Scopes requested:** email, profile (display name + photo)
- **First-time path:** Google → upsert user row → prompt for mobile (₹wallet ID) + DOB (for kundli) → land on Home
- **Returning path:** Google → match google_id → issue JWTs → land directly on Home
- **Astrologer-specific:** Google login allowed only after admin-approved KYC; pre-KYC users land on "Application Under Review" screen

B. Biometric Login (re-entry, secure)

After the user has logged in once via Google or OTP, they are offered biometric enrolment. Subsequent app launches verify with fingerprint / FaceID / iris, then unwrap the cached refresh token from secure storage.

- **Flutter SDK:** local_auth 2.3.x — fingerprint, FaceID, iris on Android & iOS
- **Token model:** Refresh token stored in flutter_secure_storage. Biometric only unlocks the wrapper — credentials never re-entered
- **Fallback:** If biometric fails 3× or hardware unavailable → fall back to OTP / Google flow
- **Enrolment screen:** Shown immediately after first OTP verify (User app) or first KYC approval (Astrologer app)
- **Re-prompt rule:** Always on cold start; after 30 min of background; before wallet recharge ≥ ₹1000; before withdrawal (Astrologer)
- **Why mandatory:** Wallets hold real ₹. Astrologers withdraw real ₹. Biometric makes session theft far harder than PIN.

C. Mobile + OTP (fallback / non-Google users)

- **Flow:** +91 mobile → send 6-digit OTP via MSG91 → enter OTP → server validates → issue JWTs
- **Use cases:** Users without Google account; biometric hardware unavailable; account recovery
- **Rate-limit:** 5 OTP/hour per number; 30 OTP/day per IP; bcrypt-hashed OTP at rest

5.2 Token & Session Model

Token	Lifetime	Where Stored	Rotates On
Google ID Token	1 hour	In-memory only	Each Google sign-in
Our Access JWT	15 minutes	In-memory (Bloc state)	Refresh call
Our Refresh JWT	30 days	flutter_secure_storage (biometric-wrapped)	On use (rolling)
OTP code	5 minutes	PostgreSQL (bcrypt)	New OTP request

5.3 Auth State Diagram (Every Phase)

The Universal Auth Guard

1) App launches → check flutter_secure_storage for refresh token. 2) Token exists → prompt biometric → unwrap → call /auth/refresh → land on Home. 3) Token missing or expired → land on Sign-In screen → Google / OTP path → on success, prompt to enable biometric for next time. 4) Any 401 from API → silent refresh attempt → if fails → kick to Sign-In. 5) Every screen sits behind a go_router redirect that checks the AuthCubit state.

5.4 Why Both Methods in Every Phase

- Google = friction-free onboarding & account portability (one tap on a new device).
- Biometric = friction-free daily re-entry without compromising security.
- OTP = inclusive — works for users without Google, on shared devices, in account recovery.
- Combined: highest conversion at signup AND highest security on session.
- Each phase below explicitly calls out which Auth Gate applies (signup gate / re-entry gate / high-value action gate).

6. User App — Module Breakdown

Seven modules, ~25 screens. Below: each module lists its screens, key features, and the auth touch-points.

Module 1 · Onboarding & Authentication

🔐 AUTHENTICATION GATE · Google Sign-In + Biometric Login

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Splash (1.5s logo animation with diya flicker)
- Welcome / Sign Up (Google · Apple · Mobile)
- OTP Verification (6-digit, 60s resend)
- Biometric Enrolment opt-in (immediately post-OTP)
- Language Picker (English / हिन्दी / Malayalam / Tamil / Telugu)
- Profile Completion (DOB · birth time · birth place — used later for Kundli)
- Permissions priming (notification, microphone, camera)

Key Flows

- Google Sign-In → upsert → if first time, redirect to Profile Completion → Home
- OTP path → verify → Biometric enrolment prompt → Profile Completion → Home
- Returning user (cold start) → Biometric prompt → Home (no re-entry of credentials)

Module 2 · Home & Discovery

🔐 AUTHENTICATION GATE · Google Sign-In + Biometric Login

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Home (greeting · wallet chip · promo banner · quick tools · celebrity wishes · featured astrologers)
- Search Astrologers (top-rated, price asc/desc, language filter)
- Astrologer Profile (bio · skills · languages · 3 rate cards · reviews · CTA)
- Celebrity Wishes Viewer (vertical reel)
- Notifications Center (incoming, system, promotional)

Highlights

- Featured Astrologers grid uses Redis-cached "currently online" pool — sub-second response.
- LIVE badge animates when astrologer is broadcasting; tap → enter live viewer.

- Quick Tools deep-link into Cosmic Insights (Module 4) — Daily Horoscope, Planet Position, Love Compatibility, Detailed Kundli.

Module 3 · Consultation (Chat · Call · Video)

> 🔐 **AUTHENTICATION GATE** · **Google Sign-In + Biometric Login**
>
> Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Chat thread (header with timer + cost · text bubbles · attachments · 50-msg cap)
- Audio Call (pulsing photo · timer · mute · speaker · end)
- Video Call (full-bleed remote + self PIP · mute · cam-off · speaker · end)
- Pre-Call Confirmation (rate, wallet balance check)
- Post-Call Summary (duration · cost · rate astrologer 1–5★)
- Chat History (filterable Active / Completed / Expired)

Key Logic

- Pre-call wallet check: balance must cover ≥ 5 minutes at astrologer rate — else prompt recharge.
- Wallet ticker debits every 60 seconds during call (or per message in chat).
- Auto-end when balance drops below 1 minute remaining; final 30s warning toast.
- Agora RTC token fetched fresh per call from /consultation/token endpoint.
- Chat sessions are 50-message capped or 30-minute capped, whichever first; unused balance refunded.

Module 4 · Kundli & Cosmic Insights

🔐 AUTHENTICATION GATE · Google Sign-In + Biometric Login

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Cosmic Insights Grid (6 paid tiles: Daily Horoscope ₹29, Detailed Kundli ₹99, Daily Prediction ₹29, Planet Position ₹79, Love Compatibility ₹19, Birthday Number ₹19)
- Generate Kundli (form: city · date · time · ayanamsa · language)
- Kundli Chart View (North-Indian chart SVG + planet legend + download PDF)
- Daily Horoscope Selector (zodiac picker → result)
- Love Compatibility (two-sign picker)
- Planet Position Calculator (current planetary positions for entered birth)

Backend

- Kundli calculation: Swiss Ephemeris (pyswisseph) for accurate planetary positions
- Lahiri / Raman / KP ayanamsa support
- PDF generation: WeasyPrint or ReportLab — astrologer-branded template, emailed + S3-stored

Module 5 · Live Streaming (Viewer Side)

🔐 AUTHENTICATION GATE · Google Sign-In + Biometric Login

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Live Stream Viewer (host video + viewer count + chat overlay + actions)
- Super Chat Purchase Modal (₹50 / ₹100 / ₹500 / custom)
- Gift Picker (rose, garland, diya, sacred-thread — each tied to a price)
- Following Tab (live-now astrologers you follow)

Engagement Mechanics

- Super Chats appear pinned at top of overlay for 30s — drives revenue.
- Gifts trigger full-screen animation visible to host and all viewers.
- Follow → push notification when followed astrologer goes live.
- Anonymous viewers (read-only) until login — converts to register.

Module 6 · Wallet & Payments

🔐 AUTHENTICATION GATE · Google Sign-In + Biometric Login

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Wallet Home (balance hero · quick recharge grid · custom amount · transactions feed)
- Recharge Confirmation (with Razorpay sheet)
- Payment History (All · Credits · Debits filters)
- Refund Status Screen (per chat-session unused-hold refunds)

Recharge Logic

- Quick amounts: 50, 100, 500, 1000, 1500, 2000 — "POPULAR" badge on 500 (configurable).
- Custom amount min ₹10, max ₹50,000 per transaction.
- Razorpay payment → server-side webhook verification → wallet credit (idempotent).
- GST not applicable on wallet recharge (treated as advance); GST applied on consultation invoice if business-flagged user.

> **Biometric Gate — High-Value Actions**
>
> Recharges over ₹1,000 and any future "Withdraw Refund to Bank" action require fresh biometric verification, even if the session is active. This is in addition to the cold-start biometric prompt.

Module 7 · Profile & Settings

🔐 AUTHENTICATION GATE · Google Sign-In + Biometric Login

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Profile Home (avatar · name · phone · language toggle EN/हिं)
- Edit Profile (name, email, DOB, gender, birth details)
- Chat History (links Module 3)
- Payment History (links Module 6)
- Sign Up as Astrologer (redirects to Astrologer app onboarding or download)
- Helpline & Support (customer care number, WhatsApp link, email)
- About / Terms / Privacy (static webview)
- Logout (revokes refresh token, clears secure storage)

7. Astrologer App — Module Breakdown

Six modules. Higher session frequency than user app (astrologers open it 10–30× per day) — biometric login is therefore even more prominent.

Module 1 · Onboarding & KYC

🔐 AUTHENTICATION GATE · **Google Sign-In + Biometric Login**

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Welcome / Apply as Astrologer (Google Sign-In primary)
- OTP Verification
- Biometric Enrolment (offered immediately — astrologers re-enter app many times daily)
- KYC Step 1 — Personal (name, DOB, address, ID type)
- KYC Step 2 — Professional (skills, languages, years of experience, rates)
- KYC Step 3 — Documents (PAN, Aadhaar, photo, certificate optional)
- KYC Step 4 — Bank Account (for payouts)
- Application Under Review (post-submit holding screen, push when approved/rejected)

Module 2 · Dashboard & Online Status

🔐 AUTHENTICATION GATE · **Google Sign-In + Biometric Login**

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Dashboard Home (online toggle hero · today's earnings · consultation count · rating · followers)
- Schedule (set available hours per day · holiday mode)
- Quick Actions (Go Live · View Requests · See Earnings)

Online/Offline Mechanics

- Toggle ON → publish to Redis "online_pool:{lang}:{skill}" with rate cards.
- Heartbeat ping every 30s; auto-offline if missed for 90s.
- Push notification arrives within 2s of incoming request.

Module 3 · Attend Requests

🔐 AUTHENTICATION GATE · **Google Sign-In + Biometric Login**

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Incoming Request Card (full-screen with 30s countdown · Accept · Decline)
- Waiting Queue (3+ pending requests sorted by arrival)
- Attend Chat (astrologer-side chat with live earnings ticker showing ₹ earned)
- Attend Audio Call (mirrored user-side call screen — ₹/min earning instead of cost)
- Attend Video Call

Module 4 · Go Live

🔐 AUTHENTICATION GATE · **Google Sign-In + Biometric Login**

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Pre-Live Setup (camera preview · title · camera/mic toggles · allow gifts toggle · chat overlay toggle)
- Live Broadcast (your camera · viewer count · chat overlay · super chat highlights · end button)
- Post-Live Summary (duration · peak viewers · total super-chat earnings · new followers · share replay)

Module 5 · Earnings & Payout

🔐 AUTHENTICATION GATE · **Google Sign-In + Biometric Login**

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Earnings Home (available balance hero · this-month breakdown · recent payouts feed)
- Withdraw to Bank (amount · confirm · biometric re-verify · success)
- Tax Summary / Form 16A download (annual)
- Commission Rate Info (transparent — "platform keeps X%")

Biometric Gate — Withdrawals

Every withdrawal — regardless of amount — requires fresh biometric verification at the confirmation step. This is a separate gate from the cold-start biometric. Failure to authenticate cancels the withdrawal.

Module 6 · Profile, Reviews & Support

🔐 AUTHENTICATION GATE · **Google Sign-In + Biometric Login**

Every screen in this phase respects the global auth guard. Re-entry to the app triggers biometric (FaceID / Fingerprint) verification before showing protected routes. Session token stored in flutter_secure_storage; biometric only unlocks the token wrapper — actual credentials never re-prompted.

Screens

- Profile (public-facing card · edit bio · update photo · skills · languages · rates)
- Reviews (all reviews + ability to thank user)
- Followers list
- Settings (notification prefs, language, logout)
- Support (chat with platform admin)

8. Database Schema

PostgreSQL 16 with SQLAlchemy 2.0 models. Below: core tables (truncated columns shown — full DDL in repo).

Table: users

All app users (seekers). One row per registered account.

Column	Type / Notes
id	UUID PK
google_id	TEXT UNIQUE NULL
mobile	VARCHAR(15) UNIQUE
email	VARCHAR(255) NULL
name	VARCHAR(120)
avatar_url	TEXT NULL
dob	DATE NULL
birth_time	TIME NULL
birth_place	VARCHAR(180) NULL
language	VARCHAR(20) DEFAULT 'en'
biometric_enabled	BOOLEAN DEFAULT false
wallet_balance	NUMERIC(12,2) DEFAULT 0
created_at, updated_at	TIMESTAMPTZ

Table: astrologers

Astrologer accounts (separate from users; one-to-many possible if same person is both).

Column	Type / Notes
id	UUID PK
user_id	UUID FK → users.id (optional self-link)
google_id	TEXT UNIQUE NULL
mobile	VARCHAR(15) UNIQUE
display_name	VARCHAR(120)
photo_url, bio	TEXT
skills	TEXT[] (e.g. {Vedic, Palmistry})

Column	Type / Notes
languages	TEXT[]
years_experience	SMALLINT
rate_chat_per_msg	NUMERIC(8,2)
rate_call_per_min	NUMERIC(8,2)
rate_video_per_min	NUMERIC(8,2)
kyc_status	ENUM(pending, approved, rejected)
is_online, is_live	BOOLEAN
avg_rating, review_count	NUMERIC / INT
available_balance	NUMERIC(12,2)

Table: consultations

Every chat / call / video session — even cancelled ones.

Column	Type / Notes
id	UUID PK
user_id, astrologer_id	UUID FKs
type	ENUM(chat, call, video)
rate	NUMERIC(8,2)
started_at, ended_at	TIMESTAMPTZ
duration_seconds, message_count	INT
amount_charged, commission, astrologer_payout	NUMERIC(10,2)
status	ENUM(requested, accepted, in_progress, completed, declined, cancelled, expired)
agora_channel, agora_uid_user, agora_uid_astrologer	VARCHAR
user_rating, user_review	SMALLINT, TEXT

Table: wallet_transactions

Every credit/debit to user wallet (recharge, consultation, super chat, refund).

Column	Type / Notes
id	UUID PK
user_id	UUID FK
type	ENUM(recharge, consultation, super_chat, gift, refund)
amount	NUMERIC(10,2) (signed: + credit, – debit)
balance_after	NUMERIC(12,2)
reference_id	UUID (consultation_id / razorpay_payment_id)
status	ENUM(success, pending, failed)
created_at	TIMESTAMPTZ

Table: astrologer_payouts

Withdrawals from astrologer balance to bank.

Column	Type / Notes
id, astrologer_id	UUID
amount, status	NUMERIC, ENUM(requested, processing, paid, failed)
bank_account_last4, ifsc	VARCHAR
razorpay_payout_id	VARCHAR
requested_at, paid_at	TIMESTAMPTZ
biometric_verified_at	TIMESTAMPTZ (audit trail)

Table: live_sessions

Live broadcasts by astrologers.

Column	Type / Notes
id, astrologer_id	UUID
title	TEXT
started_at, ended_at	TIMESTAMPTZ
peak_viewers, total_views	INT
super_chat_total, gift_total	NUMERIC
agora_channel	VARCHAR

Table: kundli_reports

Generated kundli PDFs (cached for re-download).

Column	Type / Notes
id, user_id	UUID
type	ENUM(detailed, daily_horoscope, planet_position, ...)
inputs_json	JSONB (birth details)
pdf_s3_key	TEXT
amount_paid	NUMERIC
generated_at, expires_at	TIMESTAMPTZ

Table: auth_sessions

Active refresh tokens (so admin can revoke remote sessions).

Column	Type / Notes
id	UUID PK
user_id / astrologer_id	UUID FK (one of)
refresh_token_hash	VARCHAR(255) (bcrypt of refresh JWT jti)
device_label	VARCHAR(120)
biometric_wrapped	BOOLEAN
created_at, last_used_at, expires_at	TIMESTAMPTZ
revoked_at	TIMESTAMPTZ NULL

9. API Endpoint Map

FastAPI base URL:

`https://api.hindustanijyotish.in/v1` (production)

9.1 Authentication

Method	Path	Description
POST	/auth/google	Body: { id_token }. Verifies via Google JWKS, upserts user, returns { access_token, refresh_token, user }.
POST	/auth/otp/request	Body: { mobile }. Sends OTP via MSG91. Rate-limited.
POST	/auth/otp/verify	Body: { mobile, otp }. Returns { access_token, refresh_token, user }.
POST	/auth/biometric/enable	Body: { device_label, public_key }. Marks current session biometric-wrapped.
POST	/auth/refresh	Body: { refresh_token }. Rotates refresh, returns new pair.
POST	/auth/logout	Revokes current refresh token; clears server session.
POST	/auth/sessions/revoke-all	Revokes every refresh token for this user (panic-button).

9.2 User

Method	Path	Description
GET	/me	Current user profile.
PATCH	/me	Update profile (name, birth details, language).
POST	/me/avatar	multipart/form-data — uploads to S3, returns URL.
DELETE	/me	Soft-delete account.

9.3 Astrologers (Discovery)

Method	Path	Description
GET	/astrologers	Query: ?online=true&language=hi&skill=vedic&sort=top_rated&page=1.
GET	/astrologers/{id}	Full astrologer profile.
GET	/astrologers/{id}/reviews	Paginated reviews.

Method	Path	Description
POST	/astrologers/{id}/follow	Follow / unfollow toggle.

9.4 Consultations

Method	Path	Description
POST	/consultations/request	Body: { astrologer_id, type, language }. Returns consultation row + push to astrologer.
POST	/consultations/{id}/accept	Astrologer side. Returns Agora token + channel.
POST	/consultations/{id}/decline	Astrologer side.
POST	/consultations/{id}/start	Marks in_progress; both parties join Agora.
POST	/consultations/{id}/heartbeat	Every 60s during call — server debits wallet, credits astrologer.
POST	/consultations/{id}/end	Finalises duration, cost, payout. Returns summary.
POST	/consultations/{id}/rate	Body: { rating, review }. User side, post-call.
POST	/consultations/{id}/messages	Chat message (text type only).
GET	/consultations/{id}/messages	Paginated message history.
GET	/me/consultations	My past consultations (filter: active / completed / expired).

9.5 Wallet & Payments

Method	Path	Description
GET	/wallet	Balance + recent transactions.
POST	/wallet/recharge/intent	Body: { amount }. Returns Razorpay order_id.
POST	/wallet/recharge/verify	Razorpay webhook verification, credits wallet (idempotent).
GET	/wallet/transactions	Paginated with type filter.

9.6 Live Streaming

Method	Path	Description
POST	/live/start	Astrologer side. Returns Agora token, channel, live_session_id.
POST	/live/{id}/end	Astrologer side. Persists peak viewers + earnings.

Method	Path	Description
GET	/live/now	Currently-live astrologers feed.
POST	/live/{id}/join	User side. Returns viewer Agora token.
POST	/live/{id}/super-chat	Body: { amount, message }. Debits wallet, credits astrologer.
POST	/live/{id}/gift	Body: { gift_id }.
GET	/live/{id}/chat	WebSocket — chat overlay messages.

9.7 Kundli & Insights

Method	Path	Description
POST	/kundli/generate	Body: { type, dob, time, place, ayanamsa, language }. Returns PDF URL.
GET	/kundli/reports	My past reports.
POST	/horoscope/daily	Body: { zodiac_sign }.
POST	/horoscope/compatibility	Body: { sign1, sign2 }.
POST	/horoscope/planet-position	Body: birth details.

9.8 Astrologer-Only

Method	Path	Description
POST	/astrologer/kyc	Multi-step KYC submission.
POST	/astrologer/online/toggle	Body: { is_online }. Updates Redis pool.
GET	/astrologer/requests/pending	Incoming queue.
GET	/astrologer/earnings	Available balance + breakdown.
POST	/astrologer/payout/request	Body: { amount, biometric_token }. Server verifies biometric attestation before processing.

10. Flutter Project Structure

Two separate Flutter projects (one per app) sharing a private pub.dev package "hj_core" for shared models, theme, and HTTP client.

Repository Layout

```
hindustani_jyotish/ ├── apps/ │   ├── user_app/ │   │   ├── lib/ │   │   │   ├──
main.dart │   │   │   ├── app/                    ← MaterialApp.router setup │   │
│   ├── router/                 ← go_router config + auth guard │   │   │   ├──
features/ │   │   │   │   ├── auth/                ← Google + biometric + OTP │   │
│   │   ├── home/ │   │   │   │   ├── discovery/           ← astrologer search &
profile │   │   │   │   ├── consultation/        ← chat / call / video │   │   │   │
├── kundli/ │   │   │   │   ├── insights/            ← horoscope, compatibility │   │
│   │   ├── live/                 ← viewer side │   │   │   ├── wallet/ │   │   │
│   │   └── profile/ │   │   │   └── core/                    ← (proxied from hj_core) │
│   ├── android/ │   │   ├── ios/ │   │   └── pubspec.yaml │   │ │   └──
astrologer_app/ │       ├── lib/ │       │   └── features/ │       │       ├── auth/ │
│       ├── dashboard/ │       │       ├── requests/ │       │       ├── consultation/
← attend side │       │       ├── live/                ← broadcaster side │       │
├── earnings/ │       │       └── profile/ │       └── pubspec.yaml │ └── packages/
└── hj_core/         ├── lib/         │   ├── theme/                   ← AppColors,
AppTextStyles         │   ├── models/                ← User, Astrologer,
Consultation          │   ├── network/               ← Dio client, interceptors
│   ├── auth/                   ← AuthCubit, BiometricService         │   ├──
storage/                ← SecureStorage wrapper         │   └── utils/         └──
pubspec.yaml
```

Per-Feature Folder Convention

```
features/auth/ ├── bloc/ │   ├── auth_cubit.dart                  ← AuthCubit (signed-
in / out / loading) │   └── auth_state.dart ├── data/ │   ├── auth_repository.dart
← talks to dio │   ├── google_auth_service.dart         ← google_sign_in wrapper │
├── biometric_service.dart          ← local_auth wrapper │   └── otp_service.dart ├──
domain/ │   ├── user_session.dart │   └── auth_failure.dart └── ui/     ├──
sign_up_screen.dart     ├── otp_screen.dart     ├── biometric_enrol_screen.dart
└── biometric_prompt_screen.dart
```

Routing (go_router) — Auth Guard Sample

```
final router = GoRouter(   initialLocation: '/splash',   refreshListenable:
GoRouterRefreshStream(authCubit.stream),   redirect: (ctx, state) {     final auth =
authCubit.state;     final loggingIn = state.matchedLocation.startsWith('/auth');      if
(auth is AuthLoading)          return '/splash';     if (auth is AuthUnauthenticated)
return loggingIn ? null : '/auth/welcome';     if (auth is AuthBiometricRequired) return
'/auth/biometric';     if (auth is AuthAuthenticated)    return loggingIn ? '/home' : null;
return null;   },   routes: [     GoRoute(path: '/splash', builder: (_, __) =>
SplashScreen()),     GoRoute(path: '/auth/welcome', builder: (_, __) => SignUpScreen()),
GoRoute(path: '/auth/otp', builder: (_, st) => OtpScreen(mobile:
st.uri.queryParameters['m']!)),     GoRoute(path: '/auth/biometric', builder: (_, __) =>
BiometricPromptScreen()),     ShellRoute(       builder: (_, __, child) => HomeShell(child:
child),       routes: [         GoRoute(path: '/home', builder: (_, __) => HomeScreen()),
GoRoute(path: '/astrologers/:id', builder: (_, st) => AstroProfile(id:
st.pathParameters['id']!)),         // ...       ],     ),   ], );
```

11. Development Phases & Timeline

Seven phases across ~20 weeks. Each phase explicitly implements or extends the Google + Biometric auth layer.

Phase 1 · Foundation & Auth Layer

Duration: 3 weeks

> 🔐 **Auth in this phase**
>
> Build the entire Google Sign-In + Biometric + OTP system end-to-end. This is the foundation every other phase depends on.

Deliverables

- Both Flutter projects scaffolded with BLoC + GetIt + go_router + dio
- hj_core shared package published privately
- FastAPI backend with /auth/* endpoints (Google verify, OTP, biometric enable, refresh, logout, revoke-all)
- PostgreSQL: users, astrologers, auth_sessions tables
- flutter_secure_storage integration with biometric wrapping
- Universal Auth Guard via go_router redirect
- Sign-Up, OTP, Biometric Enrolment, and Biometric Prompt screens in both apps
- Splash screen with auto-redirect logic

Phase 2 · User App — Core Discovery & Consultation

Duration: 4 weeks

> 🔐 **Auth in this phase**
>
> Every screen sits behind the Auth Guard from Phase 1. Pre-call confirmation re-prompts biometric for wallet check > ₹500.

Deliverables

- Home, Search, Astrologer Profile screens
- Chat consultation (Agora text + custom message endpoint)
- Audio Call (Agora RTC)
- Video Call (Agora RTC)
- Pre-call wallet validation; auto-end on low balance
- Post-call rating screen
- Chat History (Active / Completed / Expired)

Phase 3 · Kundli & Cosmic Insights

Duration: 2 weeks

> 🔐 **Auth in this phase**
>
> Auth Guard required for all paid insight generation. Wallet debit attempts run through fresh access-token validation.

Deliverables

- Cosmic Insights grid (6 paid tiles)
- Generate Kundli form + chart view with planet positions
- PDF generation (Swiss Ephemeris + WeasyPrint)
- Daily Horoscope, Planet Position, Love Compatibility, Birthday Number, Daily Prediction
- Kundli PDF stored on S3 with re-download from /kundli/reports

Phase 4 · Astrologer App — Onboarding, Dashboard, Attend

Duration: 4 weeks

> 🔐 **Auth in this phase**
>
> Astrologer-side has the same Google + Biometric + OTP triad. Biometric is enforced on cold start AND on every payout step. KYC documents stored encrypted on S3.

Deliverables

- Astrologer Sign-Up + 4-step KYC
- Admin approval workflow (admin panel basic)
- Dashboard with online toggle, stats, schedule
- Incoming Request handling (push, accept/decline)
- Attend Chat / Attend Call / Attend Video screens (mirror user-side with earning ticker)
- Heartbeat-based wallet settlement (every 60s)

Phase 5 · Live Streaming (Both Sides)

Duration: 3 weeks

> 🔐 **Auth in this phase**
>
> Anonymous viewing allowed for marketing reach; super-chat and gift purchases require Auth Guard + recent biometric.

Deliverables

- Astrologer: Pre-Live setup → Live Broadcast → Post-Live summary
- User: Live viewer with chat overlay, super-chat, gift picker
- WebSocket for chat overlay
- Super-chat and gift animations
- Follow / Unfollow + push on go-live
- Live revenue settlement to astrologer balance

Phase 6 · Wallet, Payments, Payouts

Duration: 2 weeks

> 🔐 **Auth in this phase**
>
> Recharges ≥ ₹1,000 require fresh biometric. Withdrawals always require fresh biometric.

Deliverables

- Razorpay integration (recharge intent + webhook verification)
- Wallet home with quick recharge grid + custom amount
- Payment history with filters
- Astrologer Earnings + Withdraw to Bank flow
- Background Celery worker for payout processing
- Refund handling for cancelled/unused sessions

Phase 7 · Polish, QA, App Store Launch

Duration: 2 weeks

> 🔐 **Auth in this phase**
>
> Auth gates regression-tested across cold start, background-resume, expired-token, revoked-session scenarios.

Deliverables

- Edge cases: poor network, mid-call disconnect, mid-recharge crash
- Localisation polish (EN + हिं first; Malayalam/Tamil/Telugu next)
- Accessibility (font scaling, screen reader labels)
- Crashlytics + Sentry wired
- Android Play Store + iOS App Store submission
- Soft launch with 50 invited astrologers

11.x Timeline Gantt (text)

Week	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7
1–3							
4–7							
8–9							
10–13							

Week	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7
14–16							
17–18							
19–20							

12. Third-Party Integrations

Service	Used For	Pricing Note
Google Cloud (OAuth)	Google Sign-In for both apps	Free for auth scope
Agora.io	Audio call, video call, live streaming SD-RTN	~$0.99 per 1k minutes of audio; $3.99 per 1k of HD video
Razorpay	Wallet recharge, astrologer payouts	2% per recharge transaction; ₹4 per payout
MSG91	Transactional OTP SMS	~₹0.20–0.25 per SMS
SendGrid	Receipts, Kundli PDF email, system alerts	Free up to 100/day; ₹0.18 per email after
Firebase Cloud Messaging	Push notifications (incoming request, live-start)	Free
Firebase Crashlytics	Crash reporting	Free
Sentry	Backend error tracking + performance	Self-hostable, or $26/mo SaaS
Cloudflare R2 / AWS S3	Profile photos, Kundli PDFs, KYC docs	R2: free egress; ~₹2/GB storage
Swiss Ephemeris (pyswisseph)	Planetary calculations for Kundli	Free, AGPL-licensed

13. Security & Compliance

Token Storage

- flutter_secure_storage — Android EncryptedSharedPreferences + iOS Keychain.
- Refresh token is biometric-wrapped: secure storage entry can be read only after local_auth success.
- Access token lives only in BLoC state (in-memory) — never persisted.
- On logout: secure storage cleared; refresh JWT JTI revoked server-side.

Transport

- TLS 1.2+ everywhere; HSTS enabled on api.hindustanijyotish.in.
- Certificate pinning in dio interceptor for the API host.
- Agora RTC tokens minted server-side with 1-hour validity; channel-scoped.

PII & Compliance

- KYC documents (PAN, Aadhaar) stored encrypted-at-rest on S3 with server-side encryption.
- Aadhaar masking — only last 4 digits displayed in app, never logged.
- Razorpay handles card data — never touches our servers (PCI-DSS scope minimised).
- Data Privacy: comply with DPDP Act 2023 (Indian Digital Personal Data Protection Act).
- Right to deletion: /me DELETE endpoint anonymises user record (consultations preserved with masked IDs for tax records).
- Audit logs for every wallet/payout action retained 7 years.

Anti-Fraud

- Razorpay webhook signature verification (idempotency key prevents replay).
- Server-side wallet ledger — never trust client-reported balance.
- Rate limits per user: 5 OTP/hr, 10 consultations/hr, 1 withdrawal/24h.
- Astrologer payout requires both: KYC approved + biometric attestation + 24h cooling-off on first withdrawal.

14. Deployment & Release

Mobile Apps

- Android — Play Store internal testing → closed beta (50 testers) → open beta → production.
- iOS — TestFlight internal → external testers → App Store production.
- Both apps share signing certificates within Jaikalki Technology Pvt. Ltd. developer account.
- Use flutter_launcher_icons for the icon packs already generated.
- CI: Codemagic or GitHub Actions — build on push to main; auto-deploy to Play Store internal track.

Backend

- Cloud: AWS Mumbai region (ap-south-1) or Hetzner Helsinki — pick on cost vs latency.
- Containers: Docker images per service; orchestrated with Docker Compose initially, k8s when DAU > 50k.
- Database: managed PostgreSQL (AWS RDS or Hetzner managed); daily snapshots, 14-day retention.
- CI/CD: GitHub Actions → build image → push ECR → deploy via SSH-pull on the target host.
- Zero-downtime: blue-green via nginx upstream switch.
- Backups: nightly pg_dump to S3; weekly restore drill.

Environments

Env	Purpose	URL
local	Developer machines	http://localhost:8000
dev	Always-on integration env	https://dev-api.hindustanijyotish.in
staging	Pre-prod, real data shape, isolated wallet	https://staging-api.hindustanijyotish.in
prod	Live production	https://api.hindustanijyotish.in

15. Appendix — Open Questions & Risks

Open Decisions

- Commission rate per consultation — 20% / 25% / 30%? Sliding scale by astrologer tier?
- Minimum withdrawal amount and cooling-off period for new astrologers?
- GST collection: charge inclusive or exclusive of GST on consultations?
- Refund policy: full refund on call disconnect within first 60s? Pro-rata?
- Live streaming: cap super-chat amount per user per session?
- Anonymity: allow users to consult anonymously (handle instead of real name)?
- Free trial: first 5-min call free with selected astrologers?

Known Risks

- Agora cost scales with usage — at 100k minutes/month, ~₹85k/month. Need to model revenue per minute.
- Astrologer churn: many quit after low earnings month-1. Mitigation: featured-rotation algorithm for new astrologers.
- Fake astrologers: KYC is necessary-but-not-sufficient. Add: video selfie call with admin, ID-photo match, periodic re-verification.
- Live moderation: super-chats need profanity filter; gifts need cooldown to prevent spam.
- iOS App Store: live-streaming with paid super-chats may need additional review — submit with care.
- WhatsApp competition: many astrologers already operate on WhatsApp. Differentiator: wallet auto-debit + payout + reviews + live streaming.

Out of Scope for v1

- Web platform (mobile-first; web later)
- In-app gifting between users
- Multi-language voice support (call language same as chat language only)
- Astrologer team accounts (multiple people sharing one profile)
- Subscription model (₹X/month for unlimited daily horoscope)

— End of Blueprint —

Hindustani Jyotish · हिन्दुस्तानी ज्योतिष

Prepared by Jaikalki Technology Pvt. Ltd.